

Intern Backend Developer

Vi Nguyen

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SUMMARY

Final-year Information Technology student at Posts and Telecommunications Institute of Technology (PTIT HCM), specializing in backend development with NestJS and Spring Boot. I have hands-on experience building full-stack web systems through personal and team projects, including a production-grade e-learning platform. I am passionate about systems thinking, understanding real-world trade-offs in software architecture, and growing within a professional engineering environment. Actively seeking a Backend Developer internship or fresher role to contribute to and learn from a real-world codebase.

EDUCATION

Posts and Telecommunications Institute of Technology – HCM Campus (PTIT HCM) - District 9, Ho Chi Minh City | Bachelor's Degree

| 2022 - 2027

SKILLS

- Language: Upper-intermediate English, focusing on technical reading and practical communication with real-world experience.
- Backend: NestJS (Modular architecture, Guards, Interceptors, Pipes); Spring Boot (REST API, Layered architecture); RESTful API design API documentation: Swagger / OpenAPI API testing: Postman
- Tool & Environment: VS Code (proficient with extensions ecosystem) Git / GitHub; AI-assisted development; Third-party integrations (eg: Clerk, Youtube Api, ...)
- Frontend: Next.js, React, TypeScript
- DevOps / Infrastructure: Basic Docker; CI/CD.
- Database: Database design (relational schema, normalization — basic); PostgreSQL (via Neon.tech cloud DB) SQL Server; Redis (caching, session); DB tools: pgAdmin, Neon.tech console, SQL Server Management Studio

PROJECTS

StudyMate — Learning Management System (LMS) | Full-Stack Developer | Personal Project

NestJS · Next.js 15 · React 19 · TypeScript · PostgreSQL · TypeORM · Redis · Clerk · Svix · Cloudinary · YouTube Data API v3 · Tailwind CSS 4 · Swagger

- Engineered a scalable RESTful API backend for a comprehensive e-learning platform using NestJS and TypeScript, adhering to Layered Architecture and SOLID principles.
- Implemented secure authentication and Role-Based Access Control (RBAC) using Clerk, managing complex permissions across 4 roles: Student, Instructor, Staff, and Admin; integrated Svix Webhooks for real-time user sync.
- Architected a PostgreSQL database via TypeORM handling complex relationships: multi-tier course structures, quizzes, lesson progress, and multi-layered category trees.

- Developed a complete e-commerce and financial module: shopping cart, order processing, enrollments, discount coupons, and an internal wallet/payout system for instructors.
- Built the full frontend with Next.js 15 App Router and React 19: role-based dashboards for all 4 user types, unified video player (YouTube + CDN), instant search with URL sync, and KYC onboarding flow for instructors using Tailwind CSS 4.
- Integrated Cloudinary for scalable media storage and YouTube Data API v3 for centralized video delivery; applied security best practices: rate limiting (Throttler), DTO validation, and Helmet.

GitHub (BE): github.com/DuvNguyen/studymate_backend | GitHub (FE): github.com/DuvNguyen/studymate_frontend

VietSneaker — Second-Hand Shoe Marketplace | Backend Developer | Team Project

Java · Spring Boot · Spring Security · MySQL · Redis · React · Next.js · TypeScript · Cloudinary · OAuth2 · JWT · Swagger

- Built RESTful APIs using Spring Boot with layered architecture (Controller – Service – Repository), covering full e-commerce workflows: product catalog, dynamic search/filtering, shopping cart, checkout, and order history.
- Implemented authentication and authorization using Spring Security, JWT, and OAuth2; designed RBAC with distinct roles: System Admin, Product Admin, Inventory Manager, and Customer.
- Integrated an AI-driven product recommendation engine and intelligent chatbot via OpenRouter/DeepSeek APIs; connected Cloudinary for product image management.
- Built an admin dashboard with inventory management, supplier tracking, and real-time revenue analytics (Chart.js); used JobRunr for background job scheduling.

GitHub: github.com/DuvNguyen/VietSneaker

HCMC Metro Smart Management System | Full-Stack Developer | Team Project

React 19 · TypeScript · Node.js · Express.js · PostgreSQL · TensorFlow.js · Face-api.js · Leaflet · PayOS · Swagger

- Comprehensive digital platform for metropolitan railway operations featuring real-time monitoring, automated e-ticketing, and AI-powered biometric security.
- Developed an AI face recognition authentication system using TensorFlow.js and Face-api.js, aggregating facial descriptors from multiple angles for biometric login and ticket validation.
- Implemented an automated e-ticketing system with PayOS payment gateway and node-cron for scheduled ticket expiration and status updates.
- Integrated Leaflet for interactive geospatial metro maps, enabling administrators to manage station data and monitor system status in real time.
- Built a sophisticated admin command center using Ant Design and Recharts to manage customer data, line schedules, promotional gift codes, and comprehensive system statistics.
- Implemented multi-language support (i18next) for both local and international users; applied Framer Motion animations for a polished UI experience.

GitHub (BE): github.com/Ryhtruly/metroHcmc---web-service | GitHub (FE): <https://github.com/Ryhtruly/metroHcmc--Admin>

CERTIFICATES

TOEIC 840 — IIG Vietnam, October 2025

LINKS

LinkedIn: [linkedin.com/in/nducviinformation/](https://www.linkedin.com/in/nducviinformation/)

GitHub: <https://github.com/DuvNguyen>

Portfolio: <https://duvnguyen.github.io/ducvi-portfolio/>